AIIMS CRE 2025: OT Technician Exam Prep

Complete MCQ Collection with Explanations

Q1. What is the connector size of an Endotracheal (ET) tube?

A) 10mm

B) 22mm

C) 15mm

D) 30mm

Correct Answer: C) 15mm

Explanation: The standard connector size for Endotracheal (ET) tubes is universally 15mm. This ensures compatibility with standard breathing circuits.

Q2. What is the least MAC of Sevoflurane?

A) 0.5

B) 1

C) 2

D) 3

Correct Answer: C) 2

Explanation: The Minimum Alveolar Concentration (MAC) for Sevoflurane is typically around 2% (2.05% specifically), which is used to compare the anesthetic potency.

Q3. What is the Pin Index for Oxygen (O_2)?

A) 2, 6

B) 3, 5

C) 1, 5

D) 2, 5

Correct Answer: D) 2, 5

Explanation: The Pin Index Safety System prevents the wrong gas from being connected to a machine. For Oxygen, the pins are at positions 2 and 5.

Q4. Which artery is most commonly used for invasive blood pressure monitoring?

A) Axillary artery

B) Femoral artery

C) Brachial artery

D) Radial artery

Correct Answer: D) Radial artery

Explanation: The radial artery is most commonly used for arterial lines due to its superficial location, ease of access, and the presence of collateral circulation (via the ulnar artery).

Q5. Laryngoscope blade is made up of?

A) Steel

B) Plastic

C) Wood

D) Rubber

Correct Answer: A) Steel

Explanation: Reusable laryngoscope blades are made of stainless steel because it is durable, doesn't damage easily, and can be easily sterilized.

Q6. Which of the following local anesthetics is an ester?

A) Lignocaine

B) Bupivacaine

C) Procaine

D) Ropivacaine

Correct Answer: C) Procaine

Explanation: Procaine is an ester-type local anesthetic. The others (Lignocaine, Bupivacaine, Ropivacaine) belong to the amide group of local anesthetics.

Q7. What is the minimum normal oxygen saturation (SpO_2) for a patient?

A) 90%

B) 85%

C) 95%

D) 80%

Correct Answer: C) 95%

Explanation: The normal oxygen saturation (SpO_2) for a healthy patient typically ranges from 95% to 98%.

Q8. Boyle's Law is used in anesthesia for?

A) Pulse oximeter

B) Vaporizer

C) Pressure-volume relation

D) Ventilator

Correct Answer: C) Pressure-volume relation

Explanation: Boyle's Law explains the inverse relationship between the pressure and volume of a gas at a constant temperature.

Q9. What is the primary use of soda lime in an anesthesia machine?

A) To humidify gases

B) To absorb CO_2

C) To absorb O_2

D) To regulate oxygen flow

Correct Answer: B) To absorb CO_2

Explanation: Soda lime is placed in the breathing circuit to absorb the carbon dioxide (CO_2) exhaled by the patient, allowing the safe rebreathing of anesthetic gases.

Q10. Which monitor is most sensitive in detection of hypoventilation?

A) Pulse oximeter

B) Capnography (EtCO2)

C) ECG

D) Blood pressure monitor

Correct Answer: B) Capnography (EtCO2)

Explanation: Capnography measures end-tidal CO2 (EtCO2) and provides immediate breath-to-breath feedback, making it the most sensitive monitor for early detection of hypoventilation.

Q11. Caudal anesthesia can not use in which surgery?

A) Hernia repair

B) Circumcision

C) Hydrocephalus

D) Foot surgery

Correct Answer: C) Hydrocephalus

Explanation: Caudal anesthesia is a type of epidural block used for surgeries below the umbilicus (like hernia, circumcision, foot). It cannot be used for upper body/head surgeries like hydrocephalus.

Q12. A man pointing toward a photo says, 'The woman in the photo is the daughter of my grandfather's only child.' Who is the woman to the man?

A) Mother

B) Sister

C) Daughter

D) Wife

Correct Answer: B) Sister

Explanation: The grandfather's 'only child' is either the man's mother or father. The daughter of the man's parent is his sister.

Q13. What is the function of scavenging system?

A) Delivers oxygen

B) Removes CO_2

C) Removes waste gases

D) Regulates pressure

Correct Answer: C) Removes waste gases

Explanation: The scavenging system safely collects and removes waste anesthetic gases from the operating room environment to prevent exposure to healthcare workers.

Q14. What is a common complication of spinal anesthesia?

A) Nausea

B) Hypotension

C) Headache

D) Infection

Correct Answer: B) Hypotension

Explanation: Spinal anesthesia often causes a sympathetic blockade, which leads to peripheral vasodilation and a drop in blood pressure (hypotension).

Q15. Sugammadex can not reverse which non-depolarizing agents?

A) Rocuronium

B) Vecuronium

C) Atracurium

D) None of Above

Correct Answer: C) Atracurium

Explanation: Sugammadex is specifically designed to encapsulate and reverse steroidal muscle relaxants (like Rocuronium and Vecuronium). It does not reverse benzylisoquinolines like Atracurium.

Q16. What is the best method for securing the airway?

A) Oral Airway

B) Laryngeal mask airway (LMA)

C) Endotracheal tube (ETT)

D) Nasal Airway

Correct Answer: C) Endotracheal tube (ETT)

Explanation: Endotracheal intubation is considered the gold standard for securing and maintaining a patent airway while preventing aspiration.

Q17. What is the normal respiratory rate (RR) for an adult?

A) 8-12

B) 20-30

C) 12-20

D) <20

Correct Answer: C) 12-20

Explanation: A normal resting respiratory rate for a healthy adult is typically between 12 and 20 breaths per minute.

Q18. Where is EtCO2 analyzer placed?

A) In Expiratory Limb

B) Near Mask

C) Flowmeter

D) Near APL valve

Correct Answer: B) Near Mask

Explanation: The EtCO2 analyzer (capnograph) is usually placed near the patient airway, often between the mask/endotracheal tube and the breathing circuit (Y-piece) for the most accurate reading.

Q19. What is the use of an HME filter?

- A) To deliver oxygen
- B) To humidify gases
- C) To filter bacteria
- D) To regulate pressure

Correct Answer: B) To humidify gases

Explanation: HME stands for Heat and Moisture Exchanger. Its primary function is to capture heat and moisture on exhalation and return it to the patient on inhalation, humidifying the gases.

Q20. What is the colour of the N2O cylinder in India?

- A) Blue
- B) Black
- C) Green
- D) Blue with white shoulder

Correct Answer: A) Blue

Explanation: In medical gas color-coding standards (including in India), Nitrous Oxide (N2O) cylinders are painted uniformly blue.